

Create Your First
Information
System

Database
Design I

Mapping

OR Mapping
Exam Papers

Assignment
Due

Week 7

Week 8

Week 9

Week
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Reading
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Database
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Normalisation

QA

Interview

LECTURE 7: CREATE YOUR FIRST INFORMATION SYSTEM

COMP2013J: Databases and Information Systems

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HTML BASICS

HTML BASICS

- HTML is a language for describing web pages.
- **HTML** stands for **H**yper **T**ext **M**arkup **L**anguage
- HTML is not a programming language, it is a markup language
- A markup language is a set of markup tags
- The purpose of the tags are to describe page content

w3schools.com (Tutorials and Examples)

<https://www.w3schools.com/>

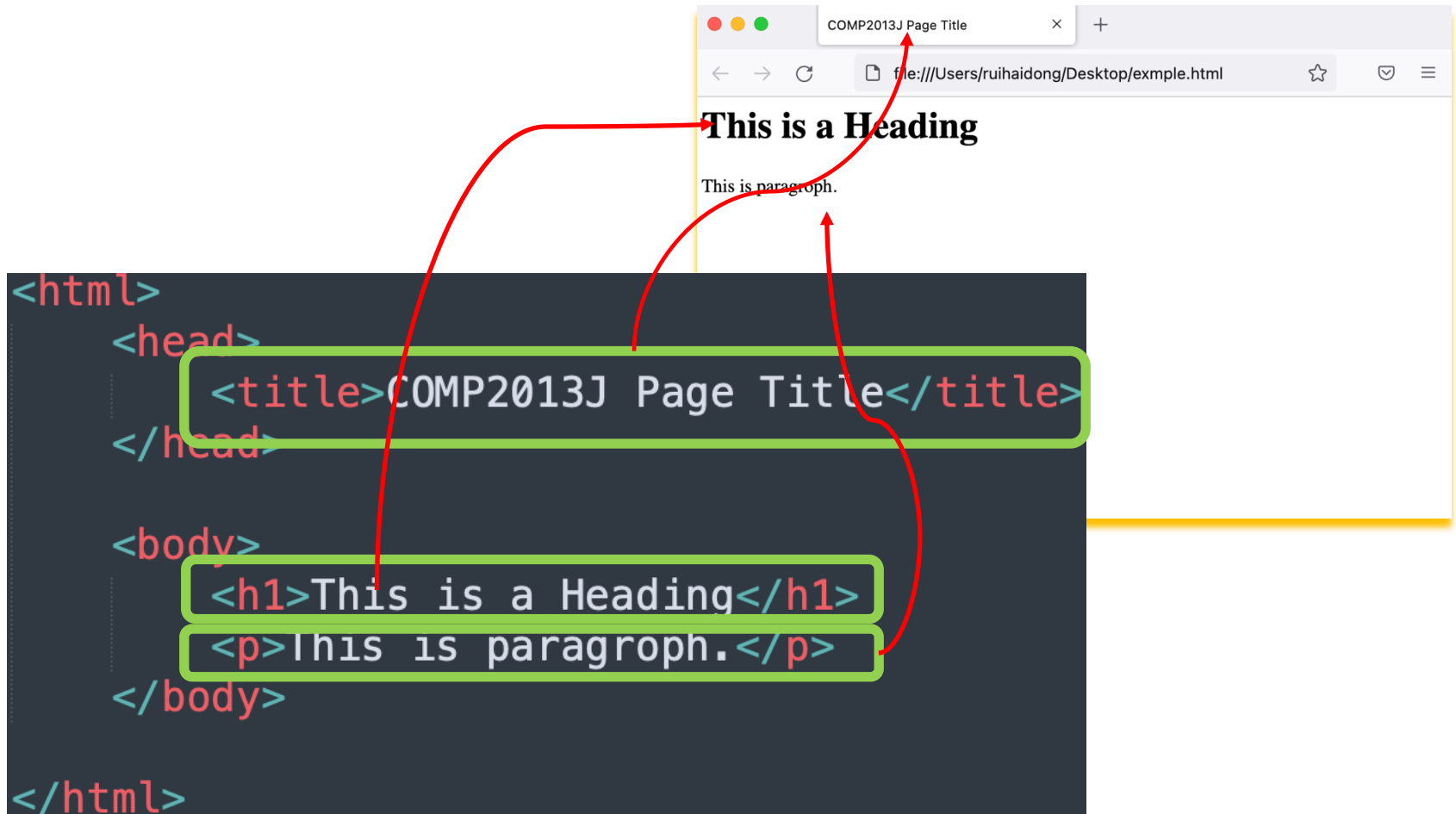
HTML Tags

- HTML markup tags are usually called HTML tags
- HTML tags are keywords (tag names) surrounded by angle brackets like `<html>` `<head>``<body>`
- HTML tags normally come in pairs like `<b>` and `</b>`
- The first tag in a pair is the start tag, the second tag is the end tag
- The end tag is written like the start tag, with a forward slash before the tag name
- Start and end tags are also called opening tags and closing tags.

HTML MAIN PART

- The text between `<html>` and `</html>` describes the web page
- The **`<head>`** elements is a container for metadata
- The following tags describe metadata:`<title>``<style>``<script>``<meta>``<link>``<base>`
- The text between **`<body>`** and **`</body>`** is the visible page content
- The text between `<h1>` and `</h1>` is displayed as a heading
- The text between `<p>` and `</p>` is displayed as a paragraph

HTML MAIN PART



TAGS AND ATTRIBUTES

Tags are sometimes enhanced by attributes, which are **name** **value** pairs that modify the tag. For example, the `img` tag (used to embed an image into a page) usually includes the following attributes:

```

```

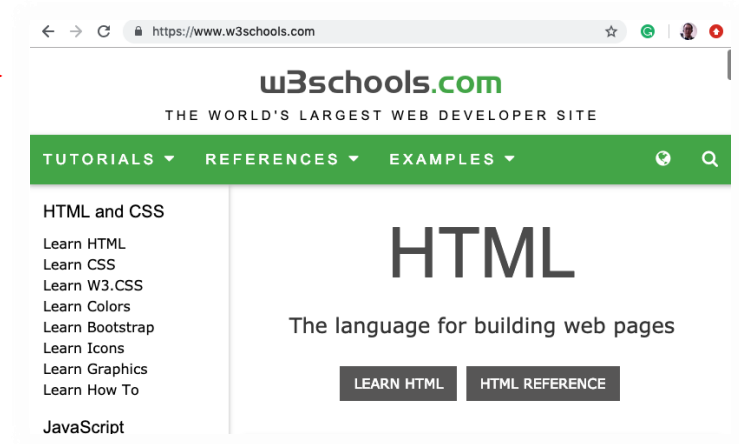
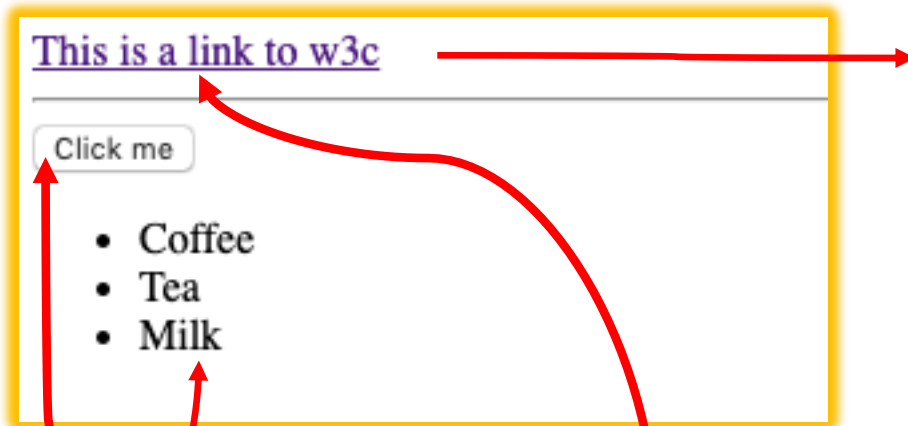
The **src** attribute describes where the image file can be found.

The **alt** attribute describes text that is displayed if the image is unavailable.

NESTED TAGS

Tags can be (and frequently are) nested inside each other.
Tags cannot overlap,
so `<a><b></a></b>` is not legal,
but `<a><b></b></a>` is fine.

LINK, BUTTON, LIST



```
<body>  
<a href="https://www.w3schools.com">This is a link to w3c </a>  
<hr/>  
<button>Click me</button>  
<ul>  
  <li>Coffee</li>  
  <li>Tea</li>  
  <li>Milk</li>  
</ul>  
</body>
```

HTML FORM

The HTML <form> element defines a form that is used to collect user input

```
<form action="/action_page.php">  
  First name:<br>  
  <input type="text" name="firstname" value="Mickey">  
  <br>  
  Last name:<br>  
  <input type="text" name="lastname" value="Mouse">  
  <br><br>  
  <input type="submit" value="Submit">  
</form>
```

defines a button for **submitting** the form data to a **form-handler**.

The form-handler is typically a server page with a script for processing input data.

The form-handler is specified in the form's **action** attribute:

First name:

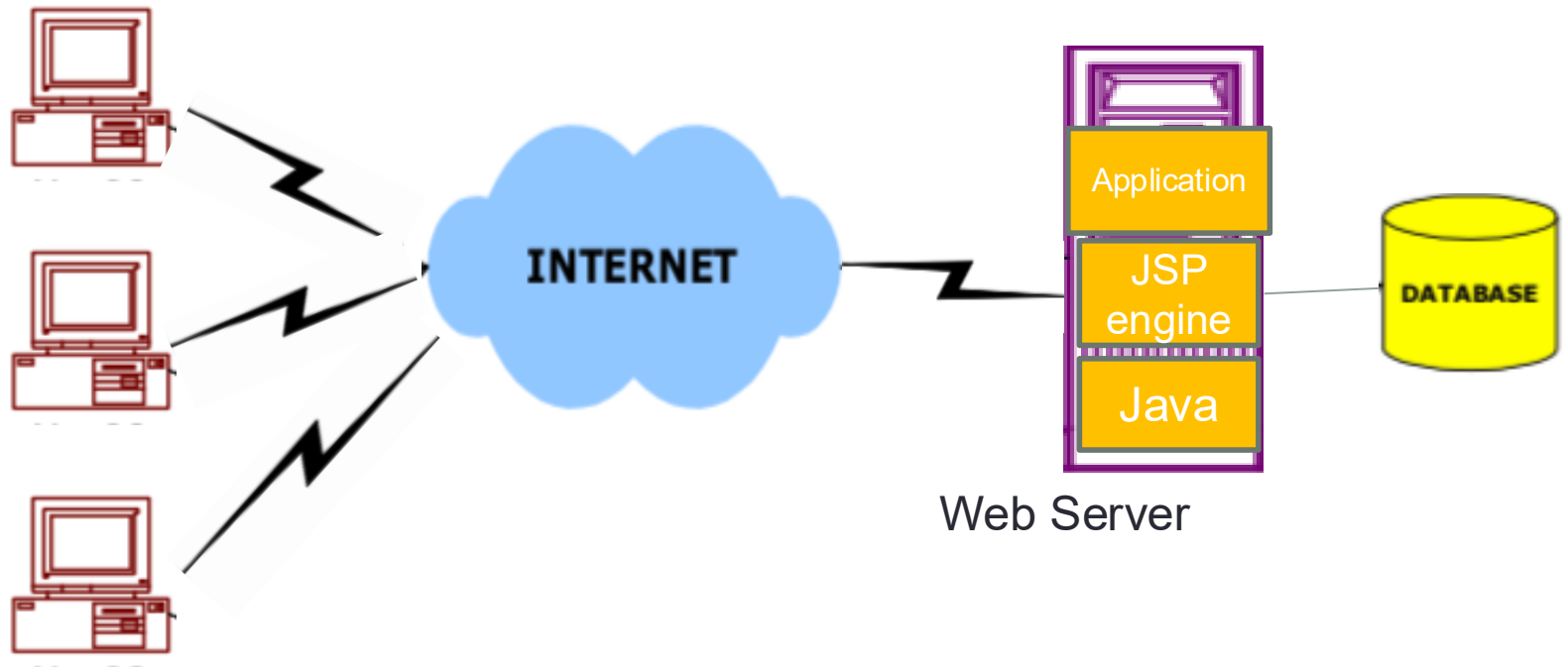
Last name:

JAVASERVER PAGES (JSP) BASICS

Introduction to JSP

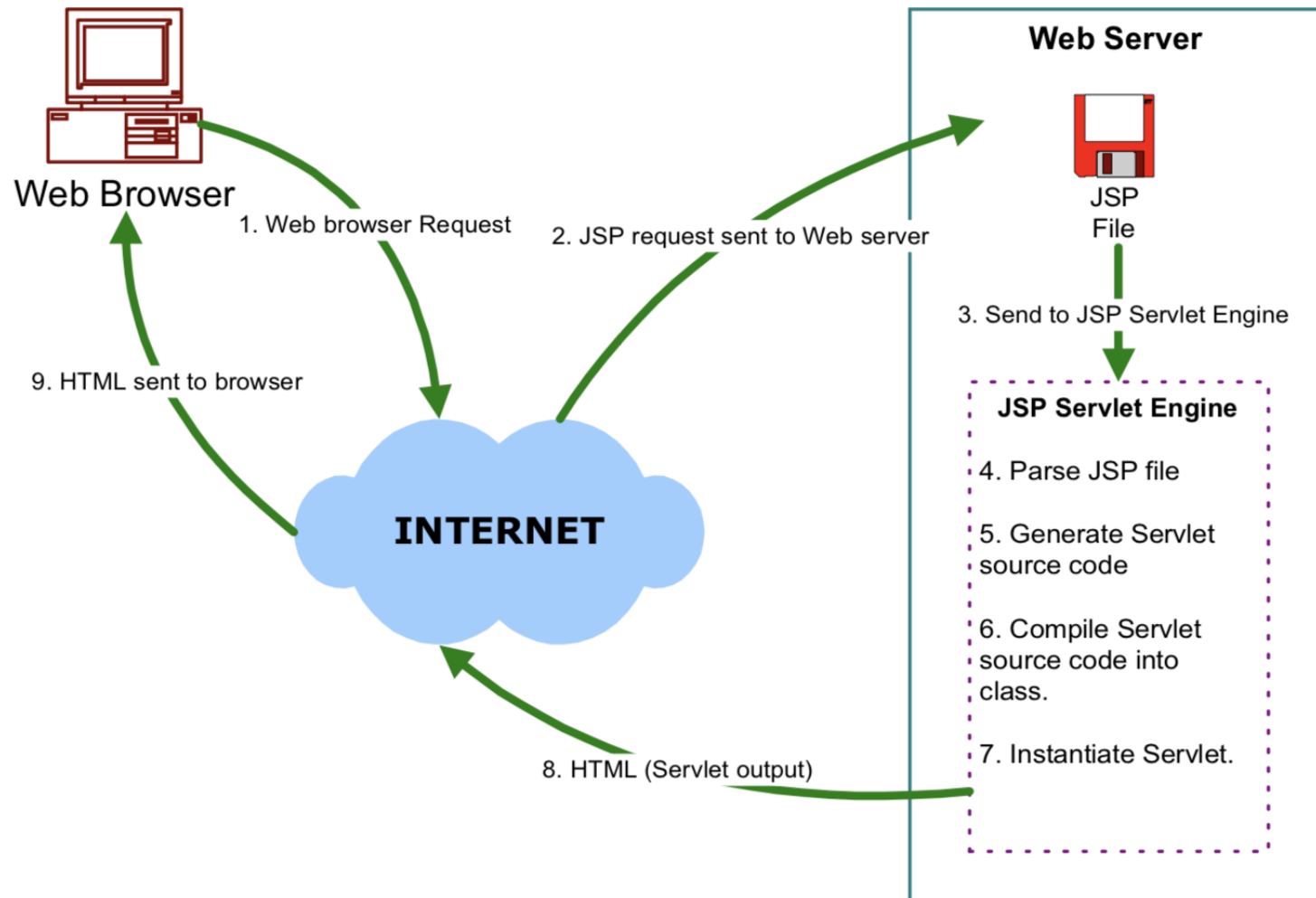
- JavaServer Pages (JSP) is a technology based on the Java language and enables the development of dynamic web sites. JSP was developed by Sun Microsystems to allow server side development. JSP files are HTML files with special Tags containing Java source code that provide the dynamic content.

Typical Structure



Different clients connecting via the Internet to a Web server.

JSP Architecture



Create your first JSP page

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<html>
<head>
<title>My First JSP Page</title>
</head>
<body>
<%
    System.out.println("Hello World");
%>
</body>
</html>
```


Scriptlet Tag <%...%>

Between <%...%> tags, any valid Java code is called a Scriptlet.

```
<%  
    String username = "ruihai";  
    System.out.println(username);  
%>
```

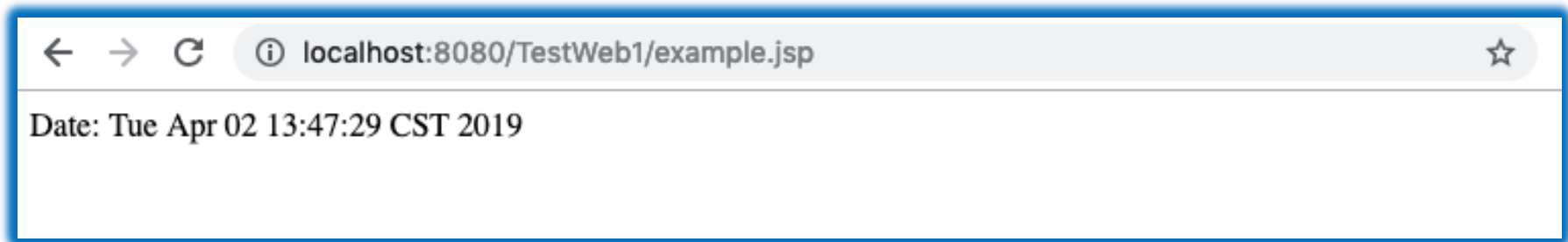
Expression Tag `<%=...%>`

This tag allows the developer to embed any Java expression and is short for `out.println()`.

A semicolon (;) does not appear at the end of the code inside the tag.

For example, to show the current data and time.

Date: `<%=new Java.util.Date()%>`



Directive Tag `<%@ directive ...%>`

A JSP directive gives special information about the page to the JSP engine.

There are three main types of directives:

- 1) **Page - processing information for this page.**
- 2) **Include – files to be included.**
- 3) **Tag library – tag library to be used in this page.**

Page Directive

language	Which language the file uses.	<code><%@ page language="java" %></code>
import	Import the classes used in the JSP	<code><%@ page import="java.util.*" %></code>

Include directive

Allows a JSP developer to include contents of a file inside another. Typically include files are used for navigation, headers and footers that are common to multiple pages.

`<%@ include file="footer.html"%>`

Implicit Objects

There are several objects that are automatically available in JSP called implicit objects.

Variable	Type
request	<code>javax.servlet.http.HttpServletRequest</code>
response	<code>javax.servlet.http.HttpServletResponse</code>
out	<code>javax.servlet.http.JspWriter</code>
session	<code>javax.servlet.http.HttpSession</code>
<code>pageContent</code>	<code>javax.servlet.jsp.PageContext</code>
<code>application</code>	<code>javax.servlet.ServletContext</code>
<code>config</code>	<code>javax.servlet.ServletConfig</code>
<code>page</code>	<code>java.lang.Object</code>
<code>exception</code>	<code>java.lang.Throwable</code>

request

Access to information associated with a request. This object is normally used in looking up parameter values and cookies.

```
<%  
    String username = request.getParameter("username");  
%>
```

A diagram illustrating a web form. It consists of three stacked light blue rectangular input fields. The top field contains the text 'ruihai' and is highlighted with a red rounded rectangle. A red arrow points from this field up and to the right towards the 'request' parameter in the code snippet above. The middle field contains six dots '.....'. The bottom field is a green rectangular button with the text 'LOGIN' in white.

index.jsp

response

The response object can be used to add or manipulate response such as redirect response to another resource, send error etc.

For example, we can redirect the response to the employeeList.jsp

```
<%  
    response.sendRedirect("employeeList.jsp");  
%>
```

session

A session object uses a key/value combination to store information.

To set a session key with a value

```
<%  
    session.setAttribute("user", u);  
%>
```

To retrieve information from a session

```
<%  
    User u2 = (User)session.getAttribute("user");  
%>
```

The return type of the method `getAttribute` is `Object`, so you will need to typecast to get the required value. If there is not a session key with that name, a null is returned.

out

The out object is used to write a message or a value to be displayed in the web page sent back to the client.

```
<%  
    String xyz = "ruihai";  
%>
```

```
<%  
    out.println(xyz);  
%>
```

```
<%=xyz%>
```


HTML and JSP Comments

HTML comments

`<!-- This is HTML comment – visible in the page source-->`

JSP comments

`<%-- This is JSP comment – not visible in the page source--%>`

JSP comments are never send to the user's browser. HTML comments are visible in the page source.

HANDS-ON YOUR FIRST INFORMATION SYSTEM

TOMCAT

<https://tomcat.apache.org/>

An open-source web server and servlet container developed by the Apache Software Foundation.



Apache Tomcat®



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Problems?

Apache Tomcat

The Apache Tomcat® software is an open source implementation of the [Jakarta Servlet](#), [Jakarta Pages](#), [Jakarta Expression Language](#), [Jakarta WebSocket](#), [Jakarta Annotations](#) and [Jakarta Authentication](#) specifications. These specifications are part of the [Jakarta EE platform](#).

The Jakarta EE platform is the evolution of the Java EE platform. Tomcat 10 and later implement specifications developed as part of Jakarta EE. Tomcat 9 and earlier implement specifications developed as part of Java EE.

The Apache Tomcat software is developed in an open and participatory environment and released under the [Apache License version 2](#). The Apache Tomcat project is intended to be a collaboration of the best-of-breed developers from around the world. We invite you to participate in this open development project. To learn more about getting involved, [click here](#).

Apache Tomcat software powers numerous large-scale, mission-critical web applications across a diverse range of industries and organizations. Some of these users and their stories are listed on the [PoweredBy](#) wiki page.

Apache Tomcat, Tomcat, Apache, the Apache feather, and the Apache Tomcat project logo are trademarks of the Apache Software Foundation.

Tomcat 10.1.39 Released

2025-03-07

The Apache Tomcat Project is proud to announce the release of version 10.1.39 of Apache Tomcat. This release implements specifications that are part of the Jakarta EE 10 platform.

Applications that run on Tomcat 9 and earlier will not run on Tomcat 10 without changes. Java EE based applications designed for Tomcat 9 and earlier may be placed in the `CATALINA_BASE/webapps-javaee` directory and Tomcat will automatically convert them to Jakarta EE and copy them to the `webapps` directory. This conversion is performed using the [Apache Tomcat migration tool for Jakarta EE tool](#) which is also available as a separate [download](#) for off-line use.

The notable changes in this release are:

- Improve the checks for exposure to and protection against CVE-2024-56337 so that reflection is not used unless required. The checks for whether the file system is case sensitive or not have been
- Use Transfer-E
- Add makensis

Full details of these c

The following documents will assist you in downloading and installing Apache Tomcat, and using many of the Apache Tomcat features.

1. [Introduction](#) - A brief, high level, overview of Apache Tomcat.
2. [Setup](#) - How to install and run Apache Tomcat on a variety of platforms.
3. [First web application](#) - An introduction to the concepts of a *web application* as defined in the Servlet Specification. Covers basic organization of the structure of a web application archive, and an introduction to the web application deployment descriptor (`/WEB-INF/web.xml`).
4. [Deployer](#) - Operating the Apache Tomcat Deployer to deploy, precompile, and validate web applications.

The Apache Tomcat Project is proud to announce the release of version 9.0.102 of Apache Tomcat. This release implements specifications that are part of the Java EE 8 platform. The notable changes compared to 9.0.100 include:

- Improve the checks for exposure to and protection against CVE-2024-56337 so that reflection is not used unless required. The checks for whether the file system is case sensitive or

Your First Information System

← → ↻ ⓘ localhost:8080/Week7Lab/index.jsp



COMP2013J Demo System

User Name:

Password:

LOGIN